

GA-Driven Recurrent Modeling for Indonesian Banking Stocks Forecasting: LSTM and Bi-LSTM with Explainable AI and Statistical Validation

Eric Julianto^{*†}, Dian Alhusari^{*}, Safrizal Ardana Ardiyansa^{*‡}

^{*}Braincore Indonesia, Indonesia

[†]Universitas Esa Unggul, Jakarta, Indonesia

[‡]Universitas Brawijaya, Malang, Indonesia

Abstract—This paper studies daily closing-price forecasting for major Indonesian banking stocks using recurrent sequence models, macroeconomic regressors, and statistical validation. The implemented ablation compares LSTM and BiLSTM under two feature settings (`Without_Macro` and `With_Macro`) and two optimization modes (default vs. budgeted tuning via Genetic Algorithm), resulting in eight experimental cases per bank. To improve rigor, we run repeated multi-seed evaluations (five seeds) across seven banks and report RMSE, MAE, MAPE, and MSE. We further apply Wilcoxon signed-rank and DM tests to verify whether performance gains are robust rather than stochastic artifacts. Empirically, BiLSTM-based configurations dominate most banks, with best-case RMSE improvements over the baseline ranging from 4.91% to 35.66% for six out of seven banks. DM-test results indicate significant forecast-error reduction in those banks, while Wilcoxon significance is limited by the small number of seeds. The study provides a reproducible benchmark and a statistically grounded foundation for testing whether macroeconomic feature engineering consistently improves forecasting.

Keywords—stock forecasting, LSTM, BiLSTM, macroeconomic features, genetic algorithm, Wilcoxon test, DM test

I. INTRODUCTION

Forecasting bank stock prices remains challenging due to non-stationarity, regime shifts, and sensitivity to macroeconomic conditions. In emerging markets such as Indonesia, banking-sector prices are influenced not only by internal price dynamics but also by external drivers including interest rates, inflation, exchange rates, and broad market movements.

Recent work in financial time series forecasting has adopted deep learning models, especially recurrent networks, to capture temporal dependencies. However, many studies still suffer from limited ablation depth and weak statistical validation. These gaps reduce confidence in whether reported improvements generalize across assets and random seeds.

This study develops a reproducible pipeline for Indonesian banking stocks that combines sequence modeling, macroeconomic regressors, repeated-seed evaluation, formal significance testing, and explainability analysis. A central question is whether feature engineering with macroeconomic variables (`With_Macro`) provides consistent gains over the baseline regime (`Without_Macro`), as often suggested in exogenous-variable forecasting literature [1], [2]. In addition, we evaluate whether model behavior remains economically plausible under Explainable AI (XAI)

inspection using SHAP-based attributions [3]. The objective is to provide a strong foundational benchmark that can be extended to more advanced architectures.

The main contributions are as follows:

- A structured ablation design over model type (LSTM vs BiLSTM), feature regime (price-only vs price+macro), and optimization mode (off vs on).
- A multi-bank, multi-seed protocol for robust performance estimation and variance analysis.
- Statistical validation using Wilcoxon signed-rank and Diebold-Mariano tests to assess significance of forecasting improvements.
- An XAI layer based on SHAP to quantify global feature relevance and compare explanation patterns across banks, feature regimes, and optimization settings.

The remainder of this paper is organized as follows. Section II reviews related literature. Section III describes the dataset and preprocessing. Section IV presents the proposed methodology. Section V details the experimental setup. Section VI discusses the results and significance tests. Section VII concludes and outlines future directions.

II. RELATED WORKS

Literature on financial time-series forecasting spans classical statistical models, recurrent architectures, and increasingly, transformer-based designs. LSTM-type models remain widely used in stock and index forecasting because they capture nonlinear temporal dependence with manageable model complexity. Recent IEEE studies continue to report competitive recurrent variants, including BiLSTM-based designs [4], [5]. Meanwhile, transformer architectures such as Temporal Fusion Transformer (TFT) [6], Informer [7], Autoformer [8], and TimeXer [1] have demonstrated strong multi-horizon performance. However, these models typically require substantially more data and compute than recurrent baselines, making LSTM-family models a practical starting point for resource-constrained settings.

Another key direction is forecasting with exogenous variables. Recent studies show that incorporating external drivers can improve prediction accuracy when those signals are informative and temporally aligned [2], [9]–[11]. This motivates our explicit comparison between `Without_Macro` and `With_Macro` settings.

Explainability is also becoming important in financial AI because predictive accuracy alone is insufficient for

risk-sensitive decisions. SHAP provides a model-agnostic additive attribution framework and has been used in stock-market factor analysis to interpret model drivers [3], [12]–[14]. Recent IEEE studies also emphasize XAI-oriented analysis for financial forecasting workflows, including LRP- and dashboard-based explainability settings [15]–[18]. Compared with black-box reporting, SHAP-based summaries can reveal whether the dominant drivers are stable across assets, seeds, and modeling regimes.

Beyond architecture and features, methodological rigor remains a common weakness in applied financial forecasting papers. Many studies report single-run metrics without repeated-seed robustness checks or formal significance testing. This paper addresses that gap by combining controlled ablation, multi-seed evaluation, and paired statistical tests (Wilcoxon and Diebold-Mariano) in one reproducible pipeline for Indonesian banking stocks.

III. DATASET AND PREPROCESSING

A. Dataset Description

This study focuses on seven Indonesian banking stocks: BBKA, BBNI, BBRI, BBTN, BDMN, BMRI, and BNGA. Daily market data include price and valuation-related variables, while macroeconomic variables include inflation-related indicators, policy rate, exchange rate, and market index features.

The target variable is the daily closing price. The analysis period follows a chronological split to avoid data leakage.

B. Feature Selection

The baseline feature set (*Without_Macro*) includes market micro and technical variables available in the bank-level dataset. The macro-augmented regime (*With_Macro*) extends this set with external variables:

- inflation rate and CPI-derived indicators,
- BI policy rate,
- USD/IDR exchange rate,
- IHSG index level and return proxy.

The final feature schema is fixed per regime before model training.

C. XAI-Oriented Feature Schema

To support consistent SHAP analysis across all experiments, feature names are kept identical across banks and seeds, and the same temporal windowing process is used in every run. The *Without_Macro* regime uses 10 market/valuation features, while *With_Macro* adds 6 external variables (total 16 features). This fixed schema allows direct aggregation of feature attribution statistics across the full ablation matrix.

D. Data Preprocessing

We apply chronological sorting, forward-fill for missing observations, and standard scaling for both features and target during model training. Sequence windows are then constructed with a fixed lookback length.

Time split protocol:

- Training: 2014-01-02 to 2019-12-31
- Validation: 2020-01-02 to 2021-12-31

- Testing: 2022-01-02 to 2024-12-30

This split reflects strict temporal validation and supports fair out-of-sample evaluation.

IV. PROPOSED METHODOLOGY

A. Overall Framework

The pipeline consists of five stages: data loading and scaling, sequence generation, model training, out-of-sample forecasting, and interpretability analysis. We evaluate multiple design factors using controlled ablation to isolate contribution from architecture, macro features, and optimization. Figure 1 summarizes the end-to-end workflow used in this study.

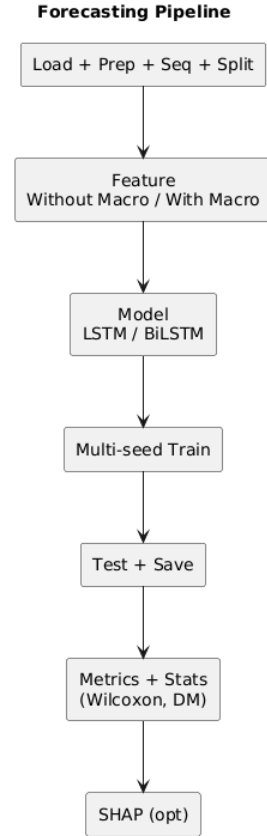


Fig. 1. Compact vertical pipeline overview used for the ablation and evaluation workflow.

B. Forecasting Models

1) *LSTM and BiLSTM*: Both models use a single-layer recurrent encoder followed by a fully connected prediction head mapping the final hidden state to a scalar forecast. The shared architectural design ensures that performance differences are attributable to the directionality of temporal encoding rather than depth or capacity changes. The two variants are:

- **LSTM**: unidirectional temporal encoder with hidden size h , producing a single forward hidden state of dimension h .
- **BiLSTM**: bidirectional temporal encoder with hidden size h , concatenating forward and backward states to produce an output of dimension $2h$.

Both variants apply a dropout rate of 0.2 between the recurrent layer and the prediction head to mitigate overfitting. The prediction head consists of a single linear layer with ReLU activation, outputting the next-day closing price.

C. Budgeted Hyperparameter Optimization via Genetic Algorithm

To isolate the effect of hyperparameter tuning from architectural choices, we introduce a constrained search procedure based on a Genetic Algorithm (GA). The GA operates on a fixed computational budget and optimizes three hyperparameters using validation loss as the fitness objective:

- **Hidden dimension** (h): searched over $\{32, 64, 128, 256\}$,
- **Learning rate** (α): searched over $\{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}\}$,
- **Sequence length** (L): searched over $\{10, 20, 30, 60\}$.

Each GA individual encodes one (h, α, L) triplet. The population size is set to 8 and the number of generations is fixed at 5, yielding a total evaluation budget of $8 \times 5 = 40$ configurations per case. Tournament selection (size 3) is used for parent selection, and single-point crossover with a per-gene mutation probability of 0.2 is applied. The hand-picked default configuration used in non-GA cases is $(h = 64, \alpha = 10^{-3}, L = 20)$. Configurations tagged as “_ga” in the results tables indicate the GA-optimized setting. This design tests whether budgeted tuning improves accuracy and stability versus fixed defaults, acting as an ablation factor for optimization effort rather than a contribution claim about the search algorithm itself.

D. Statistical Validation

Beyond average error reporting, we apply two complementary tests:

- **Wilcoxon signed-rank test**: a nonparametric paired test applied on seed-level RMSE vectors ($n = 5$) to assess whether median performance differences are significant without assuming normality.
- **Diebold-Mariano (DM) test**: applied on aligned forecast-error residuals to evaluate whether one model produces significantly lower prediction errors than another.

The Wilcoxon test provides seed-level robustness evidence, while the DM test assesses time-series-level significance. Agreement between both tests strengthens confidence in reported gains.

V. EXPERIMENTAL SETUP

A. Evaluation Metrics

Primary metrics are MSE, RMSE, MAE, and MAPE on the test partition. We report both point estimates (mean) and seed-wise variability (standard deviation) across five independent runs to characterize stability.

TABLE I
BEST-PERFORMING CASE PER BANK (MEAN $\pm$ STD OVER 5 SEEDS).

Bank	Best case	RMSE	MAE	MAPE (%)
BBCA	<code>bilstm_without_macro</code>	1881.53 $\pm$ 199.49	1721.55	18.36
BBNI	<code>bilstm_without_macro</code>	335.85 $\pm$ 57.51	252.66	4.90
BBRI	<code>bilstm_without_macro</code>	525.70 $\pm$ 123.24	435.44	8.34
BBTN	<code>bilstm_without_macro_ga</code>	51.44 $\pm$ 16.04	43.48	3.23
BDMN	<code>lstm_without_macro</code>	120.72 $\pm$ 21.92	98.32	3.74
BMRI	<code>bilstm_with_macro_ga</code>	1388.71 $\pm$ 257.96	1165.79	19.09
BNGA	<code>bilstm_without_macro</code>	244.14 $\pm$ 28.12	192.77	11.23

B. Ablation Matrix

The main ablation factors are:

- **Model family**: LSTM and BiLSTM
- **Feature regime**: `Without_Macro` (10 features) vs `With_Macro` (16 features)
- **Optimization mode**: default ($h=64, \alpha=10^{-3}, L=20$) vs GA-optimized (“_ga”)

This yields $2 \times 2 \times 2 = 8$ recurrent-model cases per bank for controlled comparison.

C. Multi-Seed Protocol

Each configuration is executed with five seeds: $\{42, 52, 62, 72, 82\}$. The complete run table stores configuration metadata, per-epoch metrics, and file paths for downstream significance analysis. All cases use identical data splits and seed sets to ensure fairness.

D. Implementation Details

Implementation is based on Python 3.10 and PyTorch 2.1. Training uses the Adam optimizer with ReduceLROnPlateau scheduling (factor 0.5, patience 5) and early stopping with validation monitoring (patience 10 epochs). The batch size is 32, and the maximum training budget is 100 epochs with a forecasting horizon of one day. All code, configuration files, and trained checkpoints are available at <https://github.com/algonacci/from-text-to-sql-agent> for full reproducibility.

E. Cross-Model Fairness Protocol

To maintain fairness, all cases use identical data splits, seed sets, sequence lengths, and optimization budgets. This avoids overstating gains from unequal search effort.

VI. RESULTS AND DISCUSSION

A. Performance Comparison Across Configurations

Table I reports the best case per bank based on mean RMSE across five seeds. BiLSTM-based settings are dominant in six out of seven banks. The best configurations are mostly `bilstm_without_macro`, with two bank-specific exceptions: `bilstm_without_macro_ga` for BBTN and `bilstm_with_macro_ga` for BMRI.

Across all eight ablation cases, the best overall average RMSE is obtained by `bilstm_without_macro_ga` (680.83), followed closely by `bilstm_without_macro` (681.26), while `lstm_without_macro` is third (741.52). In this dataset, moving from `Without_Macro` to `With_Macro` is not uniformly beneficial, and improvements are bank-dependent.

TABLE II
BEST CASE VS BASELINE (LSTM_WITHOUT_MACRO) SIGNIFICANCE SUMMARY.

Bank	Best case	Improve (%)	Wilcoxon p	DM p
BBCA	bilstm_without_macro	8.76	0.4375	2.28×10^{-13}
BBNI	bilstm_without_macro	10.47	0.0625	4.84×10^{-11}
BBRI	bilstm_without_macro	28.23	0.3125	$< 10^{-16}$
BBTN	bilstm_without_macro_ga	35.66	0.3125	2.22×10^{-16}
BDMN	lstm_without_macro	0.00	1.0000	n/a
BMRI	bilstm_with_macro_ga	11.18	0.4375	2.23×10^{-11}
BNGA	bilstm_without_macro	4.91	0.3125	4.78×10^{-5}

TABLE III
OVERALL RANKING ACROSS ALL BANKS (LOWER IS BETTER).

Case	RMSE	MAE	MAPE (%)
bilstm_without_macro_ga	680.83	591.04	10.65
bilstm_without_macro	681.26	589.76	10.74
lstm_without_macro	741.52	644.21	11.44
lstm_without_macro_ga	781.95	688.86	12.66
bilstm_with_macro_ga	806.40	722.52	13.76
bilstm_with_macro	814.04	720.31	13.14
lstm_with_macro	851.89	759.88	13.96
lstm_with_macro_ga	938.15	842.31	17.48

B. Statistical Significance Analysis

Table II compares each bank’s best case against lstm_without_macro as baseline. RMSE gains are positive in six banks, ranging from 4.91% to 35.66%.

C. Global Ranking and Regime Matrix

Table III presents an overall ranking across all banks using mean RMSE, MAE, and MAPE. The top two configurations are both BiLSTM-Without_Macro variants, while all With_Macro variants rank below them in the aggregate view.

To directly answer whether macroeconomic feature engineering improves accuracy, Table IV reports the RMSE change from Without_Macro to With_Macro per bank. Positive values indicate degradation, and negative values indicate improvement.

This evidence suggests that macro variables are useful only in specific regimes/banks rather than universally. The large RMSE spikes observed for BDMN (+202.86% for LSTM, +126.82% for BiLSTM) are consistent with temporal misalignment between macro indicators and this bank’s price dynamics: BDMN is a mid-cap bank whose microstructure-driven price behavior may respond to macro signals with a different lag structure than the zero-lag ingestion used here. This pattern is compatible with the broader literature: methods designed for exogenous variables can improve forecasting when the external signals are informative and properly aligned [1], [9], and multiple IEEE studies also report gains from macro-integrated forecasting pipelines [2], [10], [11]. In our setting, the net effect is mixed, indicating that feature relevance and temporal alignment dominate over simple feature expansion.

D. XAI Analysis with SHAP

We generated SHAP outputs for all 280 trained checkpoints (7 banks $\times$ 8 cases $\times$ 5 seeds). This produced 3,640

TABLE IV
RMSE CHANGE (%) FROM WITHOUT_MACRO TO WITH_MACRO.

Bank	LSTM delta (%)	BiLSTM delta (%)
BBCA	+20.34	+27.87
BBNI	+30.70	+39.21
BBRI	-1.27	+23.19
BBTN	-5.04	-17.72
BDMN	+202.86	+126.82
BMRI	-1.76	-4.85
BNGA	+13.25	+15.45
Mean	+37.01	+29.99

TABLE V
GLOBAL SHAP SUMMARY (ROBUST AGGREGATE OVER ALL 280 RUNS).

Feature	Mean —SHAP— (cap99)	Top-1 count
Low Price	0.00465	39
Bid Price	0.00428	31
Price Change 1 Day Percent	0.00407	28
High Price	0.00394	25
Moving Average 20 Day	0.00383	25
Price to Book Ratio	0.00383	25
Price Earnings Ratio	0.00365	20
Price to Sales Ratio	0.00362	28

feature-attribution rows over 16 unique features. Using global top-1 frequency, the most frequently dominant drivers are *Low Price* (39 runs), *Bid Price* (31), *Price Change 1 Day Percent* (28), and *Price to Sales Ratio* (28). The pattern indicates that short-horizon market microstructure and valuation factors dominate the model explanations more often than macro indicators.

Because a small number of runs showed extreme SHAP magnitudes (maximum $\approx 1.51 \times 10^{13}$), we also report a robust global ranking using a 99th-percentile cap. Under this robust aggregation, the top features remain stable: *Low Price* (0.00465), *Bid Price* (0.00428), *Price Change 1 Day Percent* (0.00407), *High Price* (0.00394), and *Moving Average 20 Day* (0.00383). Hence, the main interpretation does not depend on a few outlier explanations.

At the bank level, the most frequent top-1 SHAP feature differs across issuers: BBCA and BBNI are led by *Bid Price*, BBRI by *Price Change 1 Day Percent*, BBTN and BMRI by *High Price*, BDMN by *Price to Sales Ratio*, and BNGA by *Low Price*. This heterogeneity supports the earlier finding that forecasting behavior is regime- and bank-specific. In macro-enabled cases, *Price Change 1 Day Percent* and *Low Price* remain dominant, while macro variables (e.g., CPI index) appear as top-1 only in a subset of runs.

E. Discussion and Threats to Validity

Three validity points are important. First, the chronological split is fixed; different split boundaries may change relative ranking. Second, seed-level Wilcoxon tests are underpowered with $n = 5$, which explains why several improvements with very small DM p -values are not mirrored by Wilcoxon significance. Third, optimization benefits are

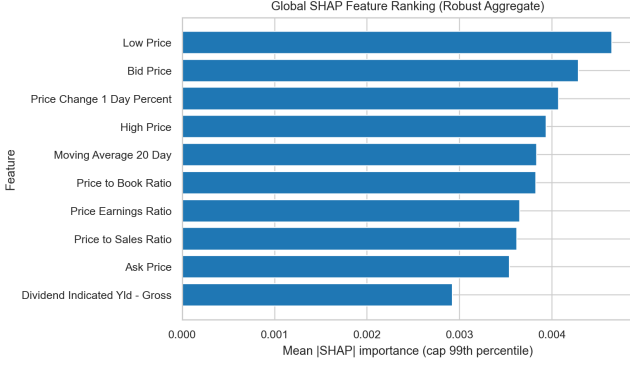


Fig. 2. Global SHAP ranking using robust mean importance (99th-percentile capped).

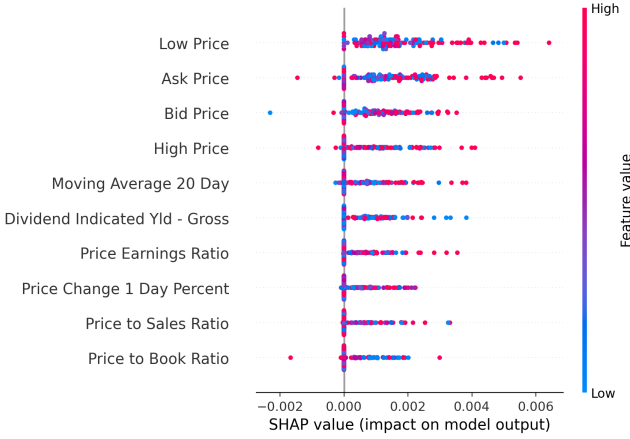


Fig. 3. SHAP beeswarm plot for the best-performing checkpoint (BBTN, BiLSTM_Data_GA, seed 62).

heterogeneous across banks, indicating risk of overfitting when tuning is not controlled.

Future work will extend this benchmark through walk-forward validation, larger seed counts, and broader stress-testing across additional market regimes for stronger temporal robustness.

VII. CONCLUSION

This paper presents a reproducible framework for forecasting Indonesian bank stock prices using LSTM-family models, macroeconomic features, multi-seed evaluation, and statistical significance testing.

Empirical results show that BiLSTM-based settings are generally stronger than LSTM across the evaluated banks, with best-case RMSE gains over the `lstm_data` baseline reaching up to 35.66%. Diebold-Mariano tests confirm significant forecast-error reduction in six banks, while Wilcoxon results are less conclusive due to limited seed count. The evidence also indicates that macro features and optimization are not uniformly beneficial, motivating bank-specific model selection rather than one-size-fits-all deployment.

Several limitations should be noted. First, the evaluation is restricted to seven Indonesian banking stocks, which constrains generalizability to other markets and sectors. Second, the five-seed protocol limits statistical power for nonparametric tests. Third, macroeconomic features were

ingested without temporal lag alignment, which may contribute to the performance degradation observed in banks such as BDMN. The framework is, however, designed for extensibility to address these constraints.

Immediate future work includes: (i) extending validation to walk-forward protocols, (ii) increasing seed count for stronger nonparametric testing power, (iii) ablation on feature lag (e.g., 7- and 30-day offsets) to investigate whether temporal misalignment explains macro feature failures, and (iv) stress-testing recurrent models under additional market regimes for broader temporal robustness.

REFERENCES

- [1] Y. Wang, H. Wu, J. Dong, G. Qin, H. Zhang, Y. Liu, Y. Qiu, J. Wang, and M. Long, "Timexer: Empowering transformers for time series forecasting with exogenous variables," *arXiv preprint arXiv:2402.19072*, 2024. [Online]. Available: <https://arxiv.org/abs/2402.19072>
- [2] S. F. Adyatma and A. Alamsyah, "The indonesia stock exchange composite prediction based on macroeconomic indicators using arima, lstm, and ann," in *2022 8th International Conference on Science and Technology (ICST)*, 2022.
- [3] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html
- [4] Y. Yang and J. Xu, "Stock index prediction based on cnn-bilstm-attention," in *2024 IEEE 2nd International Conference on Electrical, Automation and Computer Engineering (ICEACE)*, 2024.
- [5] Y. Saleh, M. Pradana, and B. R. Kartawinata, "Comparative forecasting of indonesian blue-chip stocks using arima, ann and lstm models," in *2025 IEEE 16th Control and System Graduate Research Colloquium (ICSGRC)*, 2025.
- [6] B. Lim, S. O. Arik, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," *arXiv preprint arXiv:1912.09363*, 2019. [Online]. Available: <https://arxiv.org/abs/1912.09363>
- [7] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informer: Beyond efficient transformer for long sequence time-series forecasting," *arXiv preprint arXiv:2012.07436*, 2020. [Online]. Available: <https://arxiv.org/abs/2012.07436>
- [8] H. Wu, J. Xu, J. Wang, and M. Long, "Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting," *arXiv preprint arXiv:2106.13008*, 2021. [Online]. Available: <https://arxiv.org/abs/2106.13008>
- [9] X. Qiu, Y. Zhu, Z. Li, X. Wu, B. Yang, and J. Hu, "Dag: A dual correlation network for time series forecasting with exogenous variables," *arXiv preprint arXiv:2509.14933*, 2025. [Online]. Available: <https://arxiv.org/abs/2509.14933>
- [10] S. S. Begum, N. S. Fatima, and M. Abbas, "Enhancing stock market prediction with an lstm-attention model integrating macroeconomic and sentiment features," in *2024 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)*, 2024.
- [11] J. Dechjijaruwate and C. Pisarn, "Forecasting stock prices of commercial banks in the stock exchange of thailand using machine learning," in *2025 20th International Joint Symposium on Artificial Intelligence and Natural Language Processing (ISAI-NLP)*, 2025.
- [12] C. M. Shih and K. C. Yang, "Using lightgbm + shap to analyze the impact of factors on stock market investment," in *2024 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)*, 2024.
- [13] M. H. Begum, F. M. A. Rahman, M. A. K. Azad, C. R. Debi, K. M. M. Uddin, and A. K. M. Masum, "Diverse minds think alike: A shap-guided rnn for stock price forecasting with smart sma optimization," in *2025 3rd International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings)*, 2025.
- [14] Y. Gupta S, P. Naresh, C. Nagesh, P. Praveen, N. Kirankumar, and K. Surendra, "An interpretable framework for stock market forecasting using long short-term memory (lstm) networks with shap - explainable ai (xai) method," in *2025 6th International Conference on Data Intelligence and Cognitive Informatics (ICDICI)*, 2025.

- [15] H. Tanveer, S. Arshad, H. Ameer, and S. Latif, "Interpretability in financial forecasting: The role of explainable ai in stock market," in *2024 14th International Conference on Software Technology and Engineering (ICSTE)*, 2024.
- [16] H. Tanveer and S. Latif, "Decoding stock market predictions: Insights from explainable ai using layer-wise relevance propagation," in *2024 26th International Multi-Topic Conference (INMIC)*, 2024.
- [17] J. Jagannathan and V. V. S. K. Goditi, "Integration of a hybrid model with xai for stock price forecasting: A dashboard-driven approach," in *2024 International Conference on IT Innovation and Knowledge Discovery (ITIKD)*, 2025.
- [18] S. Ayyappan, A. A. Gadgil, R. Gupta, H. B. Furaiji, S. Najah, and J. A. Jasim, "Time series forecasting framework using explainable deep learning for stock price prediction," in *2025 International Conference on Recent Innovation in Science Engineering and Technology (ICRISET)*, 2025.